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Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

Secure networks with firewalls NAT, port filtering, packet filtering, and other methods

Task:

❖ **1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.**

➢ **Aim**

➢ To configure Windows Defender Firewall on a Windows laptop or PC and create a new inbound firewall rule to block incoming network traffic on port 8080.

➢ **Introduction**

- A firewall is a network security system that monitors and controls incoming and outgoing network traffic according to predefined security rules.
- **Windows Defender Firewall** is a built-in security feature in Microsoft Windows. It helps protect a computer from unauthorized network access by allowing or blocking traffic based on firewall rules.

- In this practical, Windows Defender Firewall is used to create an inbound rule that blocks incoming traffic on port 8080. This helps prevent unauthorized network connections to services listening on that port.

➢ **Requirements**
➢ The following are required for this practical:
1) A Windows laptop or desktop computer.
2) Windows Defender Firewall.
3) Administrator access to the computer.
4) A keyboard and mouse.

➢ **Procedure to Create an Inbound Firewall Rule**
➢ **Step 1: Open Windows Defender Firewall**
1. Start the Windows laptop or PC.
2. Click the Start button.
3. Search for **Windows Defender Firewall with Advanced Security**.
4. Open the application.

➢ **Step 2: Open Inbound Rules**
1. In the left-side panel, select **Inbound Rules**.
2. In the right-side Actions panel, click **New Rule**.
3. The New Inbound Rule Wizard will open.

➢ **Step 3: Select the Rule Type**
1. Select the **Port** option.
2. Click **Next**.

➢ **Step 4: Select the Protocol and Port**
➢ For the first rule:

1. Select **TCP**.
2. Select **Specific local ports**.
3. Enter 8080.
4. Click **Next**.

- For the second rule, repeat the process but select **UDP** and enter 8080.
- **Step 5: Select the Action**
  1. Select **Block the connection**.
  2. Click **Next**.
- This option blocks incoming connections that match the rule.
- **Step 6: Select the Network Profiles**
  1. Select all three profiles:
     - Domain
     - Private
     - Public
  2. Click **Next**.
- This applies the rule to all three Windows network profiles.
- **Step 7: Name the Firewall Rule**
- For the TCP rule, enter:
  Block Incoming Port 8080 TCP
- For the UDP rule, enter:
  Block Incoming Port 8080 UDP
- Click **Finish** to create each rule.
- **5. Verification of Firewall Rules**
  1. Open Windows Defender Firewall with Advanced Security.
  2. Select **Inbound Rules**.
  3. Find the rules named:
     - Block Incoming Port 8080 TCP

- Block Incoming Port 8080 UDP

4. Verify that both rules are enabled.
5. Check that their Action is set to **Block**.

➢ The rules should now block incoming TCP and UDP traffic on local port 8080.

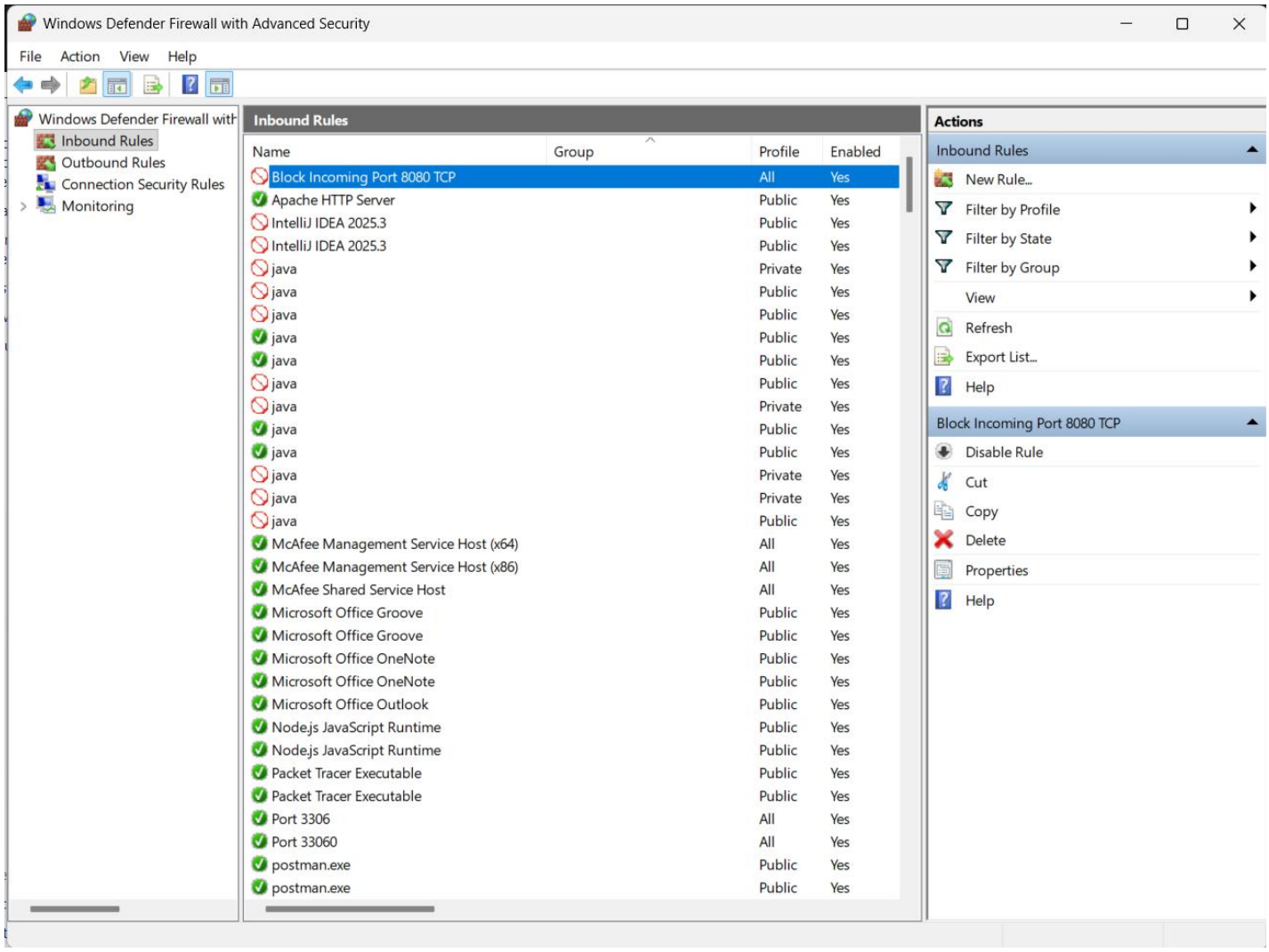
➢ **Observation**

Sr. No.	Configuration	Setting
1	Firewall	Windows Defender Firewall
2	Rule Direction	Inbound
3	Rule Type	Port
4	Protocol	TCP and UDP
5	Local Port	8080
6	Action	Block the connection
7	Network Profiles	Domain, Private, Public
8	Rule Status	Enabled

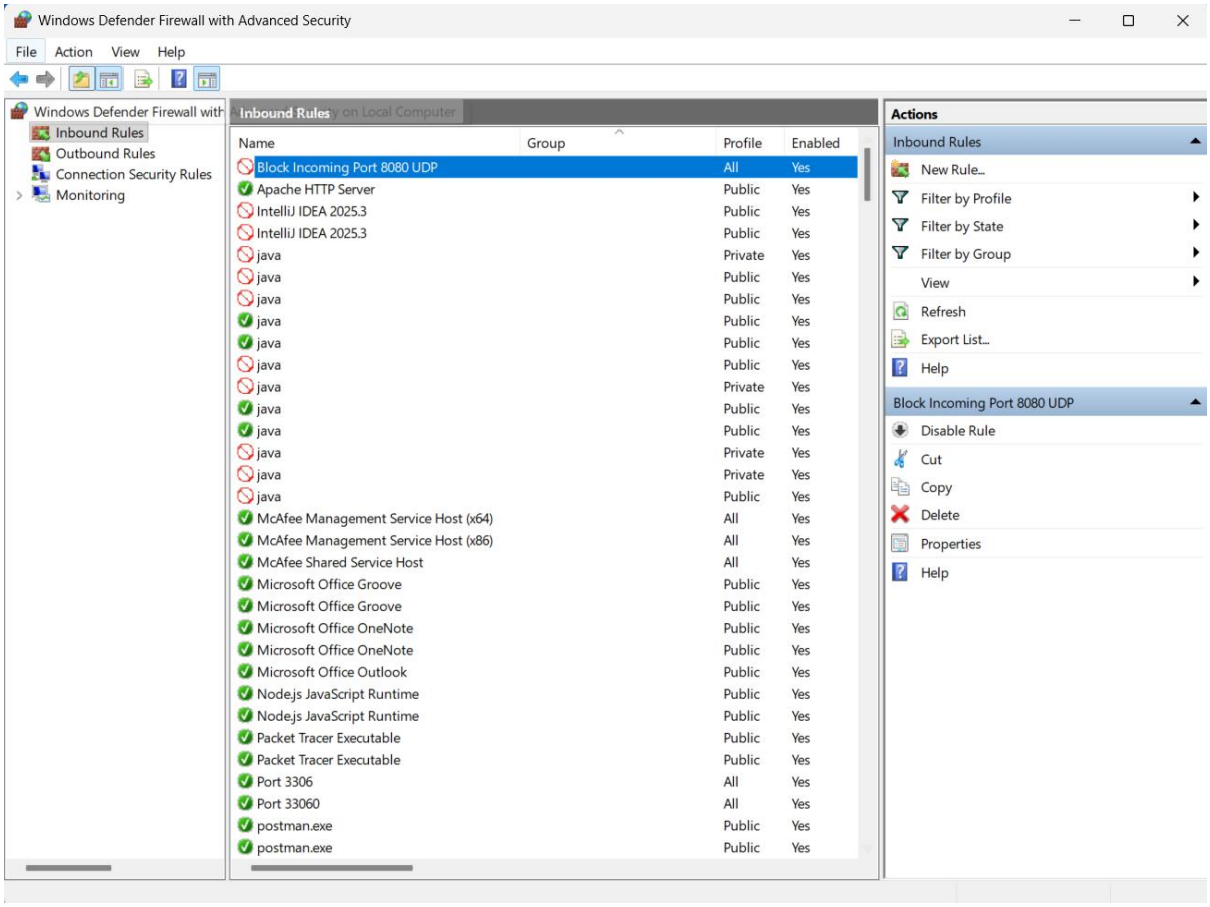
➢ **Screenshot**

➢ Take a screenshot of the **Inbound Rules** window showing both newly created rules:

- Block Incoming Port 8080 TCP



- Block Incoming Port 8080 UDP



- The screenshot should clearly display the rule names and their enabled status.

- **Result**
- Windows Defender Firewall was successfully configured by creating inbound rules to block incoming TCP and UDP traffic on port 8080 across Domain, Private, and Public network profiles.

❖ **2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access. Use any drawing tool or even hand-draw and click a photo.**

➢ **Aim**

- To understand and demonstrate Network Address Translation (NAT) by showing how a home Wi-Fi router assigns private IP addresses to multiple devices and translates them into a single public IP address for Internet access.

➢ **Introduction**

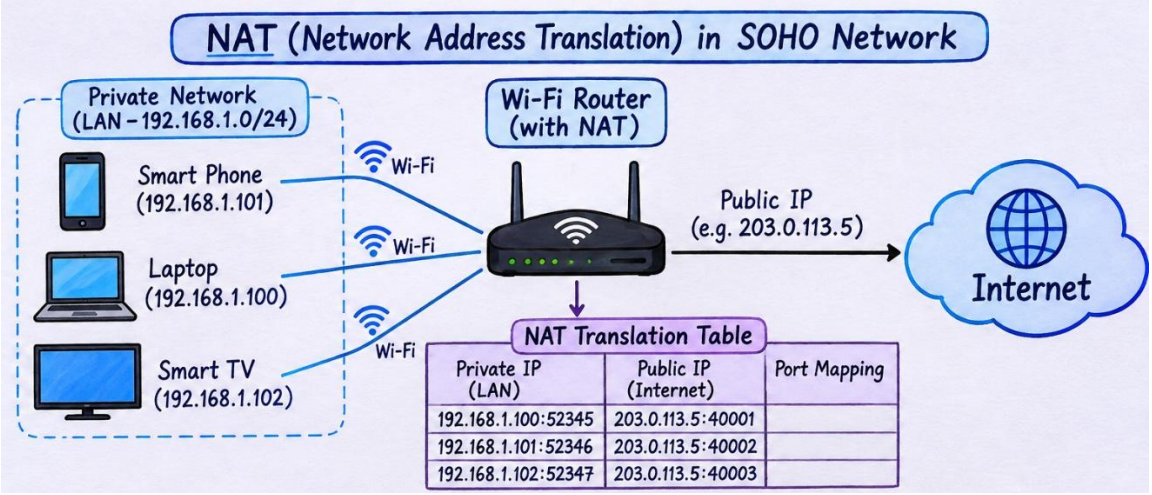
- **NAT (Network Address Translation)** is a networking technique used by routers to translate private IP addresses into public IP addresses and vice versa.
- In a home or SOHO network, multiple devices such as smartphones, laptops, and smart TVs use private IP addresses to communicate within the local network.
- When these devices access the Internet, the Wi-Fi router uses NAT to translate their private IP addresses into the public IP address assigned by the Internet Service Provider (ISP).
- This allows multiple devices to share a single public IPv4 address for Internet communication.

➢ **Requirements**

➢ The following components are required:

1. One broadband Internet connection.
2. One Wi-Fi router with NAT support.
3. One smartphone.
4. One laptop.
5. One smart TV.
6. An Internet connection provided by an ISP.

- **NAT Network Diagram**
- The following diagram illustrates how NAT works in a SOHO network.



- **IP Address Assignment Table**
- The following table shows example private IP addresses assigned to the devices and the public IP address used by the router.

Sr. No.	Device	Private IP Address	Connection Type
1	Wi-Fi Router (LAN)	192.168.1.1	Router

2	Laptop	192.168.1.100	Wireless (Wi-Fi)
3	Smartphone	192.168.1.101	Wireless (Wi-Fi)
4	Smart TV	192.168.1.102	Wireless (Wi-Fi)

- **Public IP Address of Router:** 203.0.113.5 (Example address used for documentation).

- **Working of NAT**
- NAT works through the following steps:
  1) **Private IP Assignment:** The Wi-Fi router assigns a unique private IP address to each connected device using DHCP.
  2) **Internet Request:** When a laptop, smartphone, or smart TV sends a request to the Internet, the request reaches the Wi-Fi router.
  3) **Address Translation:** The router translates the device's private IP address into its public IP address. It also tracks connection information, such as port numbers, to identify individual connections.
  4) **Internet Communication:** The translated request is sent to the Internet using the router's public IP address.
  5) **Receiving the Response:** When a response comes back from the Internet, the router checks its NAT translation table.
  6) **Forwarding to the Correct Device:** The router uses the stored connection information to forward the response to the correct device on the local network.

- **Example of NAT Translation**
- The following table shows an illustrative example of how the router maps private connections to a single public IP address.

Device	Private IP and Port	Public IP and Port
Laptop	192.168.1.100:52345	203.0.113.5:40001
Smartphone	192.168.1.101:52346	203.0.113.5:40002
Smart TV	192.168.1.102:52347	203.0.113.5:40003

- **Advantages of NAT**
  1. **Saves Public IP Addresses:** Multiple devices can share a single public IPv4 address.
  2. **Private Address Protection:** Private IP addresses are not directly exposed to external Internet networks.
  3. **Internet Sharing:** Multiple devices can access the Internet through one router.
  4. **Network Management:** NAT allows a router to manage communication between local devices and the Internet.
  5. **Improved Network Security:** NAT provides address hiding, although it does not replace a firewall or other security measures.\
- **Result**
- NAT was studied and demonstrated using a SOHO network example. A Wi-Fi router assigns private IP addresses to a smartphone, laptop, and smart TV and translates their connections to a single public IP address for Internet communication.

❖ **3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.**

➢ **Aim**

- To configure port filtering on a Wi-Fi router to block outgoing traffic on TCP port 25 (SMTP) and understand how this helps prevent spam emails from a network.

➢ **Introduction**

- Port filtering is a network security technique used to allow or block network traffic based on specific port numbers.
- Different network services use different port numbers for communication. For example, SMTP (Simple Mail Transfer Protocol) commonly uses TCP port 25 for sending email between mail servers.
- In this practical, port filtering is configured on a Wi-Fi router to block outgoing TCP traffic on port 25. This helps restrict devices on the local network from directly sending email through SMTP port 25.

➢ **Requirements**

➢ The following are required for this practical:

1. One Wi-Fi router with port filtering or outbound firewall support.
2. A laptop or computer.
3. An Internet connection.
4. A web browser.
5. Router administrator login credentials.

- **Router Configuration Details**

Configuration	Setting
Router Model	TP-Link Archer C6 (Example)
Filtering Type	Outbound Port Filtering
Protocol	TCP
Destination Port	25
Service	SMTP
Action	Deny / Block
Rule Status	Enabled

- **Step-by-Step Procedure to Configure Port Filtering**

- **Step 1: Connect to the Wi-Fi Router**
  1. Connect the laptop to the Wi-Fi router using Wi-Fi or an Ethernet cable.
  2. Open a web browser.
  3. Enter the router's administration address, such as 192.168.1.1 or 192.168.0.1.
  4. Log in using the administrator credentials.
- **Step 2: Open Firewall or Access Control Settings**
  1. Open the router's Advanced Settings.
  2. Look for the Firewall, Security, Access Control, or IP/Port Filtering section.

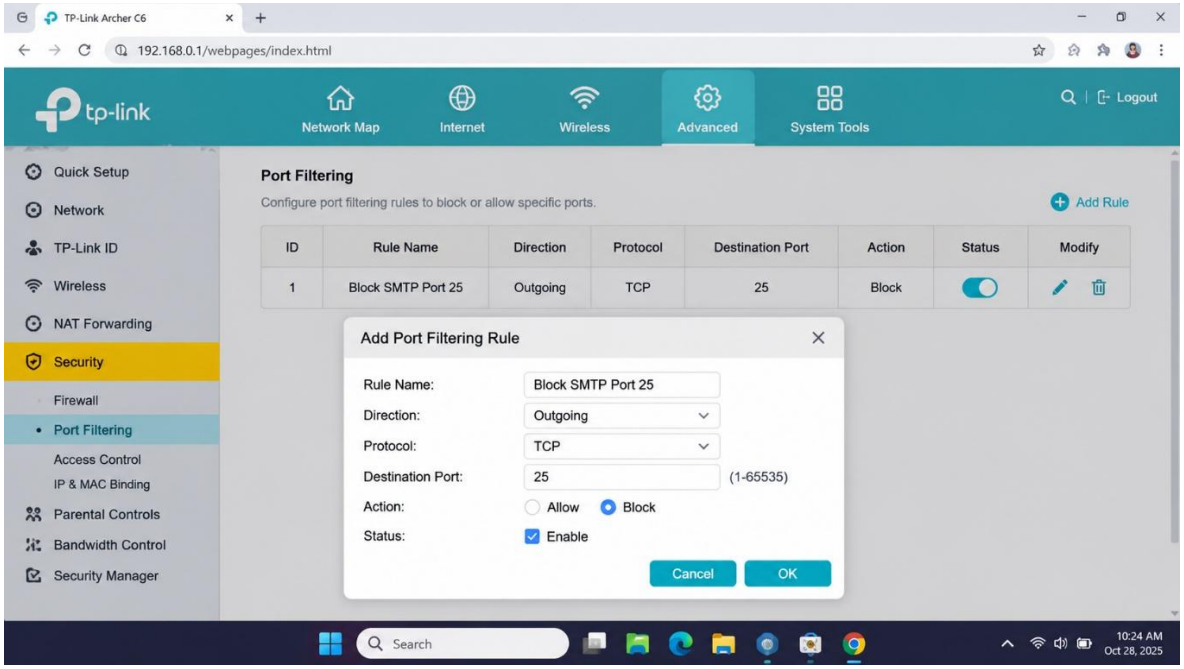
3. Open the Outbound Rules or Egress Filtering option, if available.

➢ **Step 3: Create a New Port Filtering Rule**

1. Select Add New Rule or Create Rule.
2. Enter the following configuration:

Setting	**Value**
Rule Name	Block SMTP Port 25
Direction	Outgoing
Protocol	TCP
Destination Port	25
Source	All LAN Devices
Action	Deny / Block
Status	Enabled

3. Check that the rule applies to all devices on the local network.



- **Step 4: Save and Apply the Rule**
  1. Review the rule settings.
  2. Click Save or Apply.
  3. Make sure the rule is enabled.
  4. Restart the router if the router requires it.
- **Step 5: Verify the Configuration**
  1. Open the router's Port Filtering or Firewall Rules page.
  2. Find the rule named **Block SMTP Port 25**.
  3. Verify that the protocol is TCP.
  4. Confirm that the destination port is 25.
  5. Check that the action is set to Deny or Block.
  6. Confirm that the rule is enabled.

- **Observation**

Sr. No.	Parameter	Observation
1	Firewall Feature	Port Filtering
2	Traffic Direction	Outgoing
3	Protocol	TCP
4	Port Number	25
5	Network Service	SMTP
6	Rule Action	Block
7	Rule Status	Enabled

- **How Blocking Port 25 Helps Prevent Spam**
- Blocking outgoing TCP port 25 can prevent infected or unauthorized devices on the network from directly sending large amounts of spam email through external mail servers. It reduces the misuse of the network for spam distribution, although it does not completely prevent spam because other email submission methods and ports may still be used.

- **Result**
- An outbound port filtering rule was configured to block TCP traffic on port 25 (SMTP) from devices on the SOHO network. The rule was saved, enabled, and verified in the router's firewall settings.

❖ **4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53. Focus on source/destination IPs and ports in the rule set.**

➢ **Aim**

- To examine sample network packets using packet filtering rules and identify whether each packet should be allowed or blocked based on source IP addresses, destination IP addresses, and destination port numbers.

➢ **Introduction**

- Packet filtering is a network security technique used by firewalls to control incoming and outgoing network traffic. It examines packet information such as source IP address, destination IP address, protocol, and port number.
- Based on predefined rules, a firewall decides whether to allow or block a packet. Packet filtering helps protect a network from unauthorized access and unwanted traffic.

➢ **Sample Packet Filtering Rules**

➢ Since no specific rule set was provided, the following sample rules are assumed for this assignment.

Rule No.	Source IP Address	Destination Port	Protocol	Action
1	192.168.1.0/24	80	TCP	Allow
2	10.0.0.0/8	22	TCP	Block
3	192.168.1.0/24	53	UDP	Allow
4	Any	Any	Any	Block

- **Explanation:**
  1. Allow HTTP traffic from the 192.168.1.0/24 network on TCP port 80.
  2. Block SSH traffic from the 10.0.0.0/8 network on TCP port 22.
  3. Allow DNS queries from the 192.168.1.0/24 network on UDP port 53.
  4. Block all other traffic that does not match an allowed rule.

- **Analysis of the Three Sample Packets**
- **Packet 1: HTTP Request**
  - Source IP: 192.168.1.5
  - Destination IP: 172.217.166.78
  - Destination Port: 80
  - Protocol: TCP
  - Service: HTTP
- **Decision: ALLOWED**

- **Reason:** The source IP address 192.168.1.5 belongs to the 192.168.1.0/24 network, and the destination port is 80. It matches Rule 1, which allows HTTP traffic.
- **Packet 2: SSH Request**
  - Source IP: 10.0.0.10
  - Destination IP: 34.201.23.1
  - Destination Port: 22
  - Protocol: TCP
  - Service: SSH
- **Decision: BLOCKED**
- **Reason:** The source IP address 10.0.0.10 belongs to the 10.0.0.0/8 network, and the destination port is 22. It matches Rule 2, which blocks SSH traffic.
- **Packet 3: DNS Query**
  - Source IP: 192.168.1.7
  - Destination IP: 8.8.8.8
  - Destination Port: 53
  - Protocol: UDP
  - Service: DNS
- **Decision: ALLOWED**
- **Reason:** The source IP address 192.168.1.7 belongs to the 192.168.1.0/24 network, and the destination port is 53. It matches Rule 3, which allows DNS queries.

- **Summary of Packet Filtering Results**

Packet No.	Service	Source IP	Destination IP	Destination Port	Decision
1	HTTP	192.168.1.5	172.217.166.78	80	Allowed
2	SSH	10.0.0.10	34.201.23.1	22	Blocked
3	DNS	192.168.1.7	8.8.8.8	53	Allowed

- **Result**
- Out of the three sample packets, two packets were allowed and one packet was blocked according to the assumed packet filtering rules.
  - HTTP request: Allowed
  - SSH request: Blocked
  - DNS query: Allowed

❖ 5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

➢ Aim

- To study three ways Instagram and Zomato might use firewalls and packet filtering to protect their servers from cyberattacks, unauthorized access, and misuse.

➢ Introduction

- A firewall is a network security system that monitors incoming and outgoing network traffic. It allows or blocks traffic according to predefined security rules.
- Packet filtering examines information such as source IP address, destination IP address, protocol, and port number to decide whether a network packet should be allowed or blocked.
- Large online platforms such as Instagram and Zomato handle a large amount of user information and network traffic. Therefore, firewalls and packet filtering can help protect their servers and maintain the security and availability of their services.

➢ Three Ways Instagram and Zomato Might Use Firewalls and Packet Filtering

➢ 1. Blocking Unauthorized Network Access

➢ Instagram and Zomato might use firewalls to allow only authorized network traffic to reach their servers. Packet filtering rules can allow

required services, such as secure HTTPS communication, while blocking unnecessary or unauthorized ports.

- **Example:** A firewall may allow HTTPS traffic on TCP port 443 while blocking access to unused administrative ports from the public Internet.
- **Benefit:** This reduces unauthorized access and helps protect important server resources from unwanted connections.

- **2. Protecting Servers from DDoS Attacks**
- Both platforms might use network firewalls and traffic-filtering systems to identify and limit suspicious traffic directed towards their servers.
- A Distributed Denial-of-Service (DDoS) attack attempts to make an online service unavailable by sending a large volume of unwanted traffic. Firewalls and filtering systems can help detect abnormal traffic patterns and block or limit suspicious requests.
- **Example:** If a server receives an unusually large number of requests from suspicious sources, security systems may filter or limit that traffic.
- **Benefit:** This helps maintain website and application availability for genuine users.

- **3. Network Segmentation and Monitoring**
- Instagram and Zomato might divide their infrastructure into separate network sections, such as web servers, application servers, and database servers. Firewall rules can control which sections are allowed to communicate with one another.
- They may also monitor network traffic and maintain logs to identify suspicious activities.

- **Example:** A database server may accept connections only from authorized application servers instead of allowing direct public Internet access.
- **Benefit:** This helps protect sensitive information and limits the spread of unauthorized access within the network.

- **Summary Table**

No.	Security Method	How It Helps Instagram and Zomato
1	Blocking unauthorized access	Allows required services and blocks unwanted connections.
2	DDoS protection	Helps filter suspicious traffic and maintain service availability.
3	➢ Network segmentation and monitoring	Protects databases, controls communication, and helps detect suspicious activity.

- **Result**
- Three ways in which Instagram and Zomato might use firewalls and packet filtering were studied: blocking unauthorized access, protecting servers from DDoS attacks, and using network segmentation and monitoring.